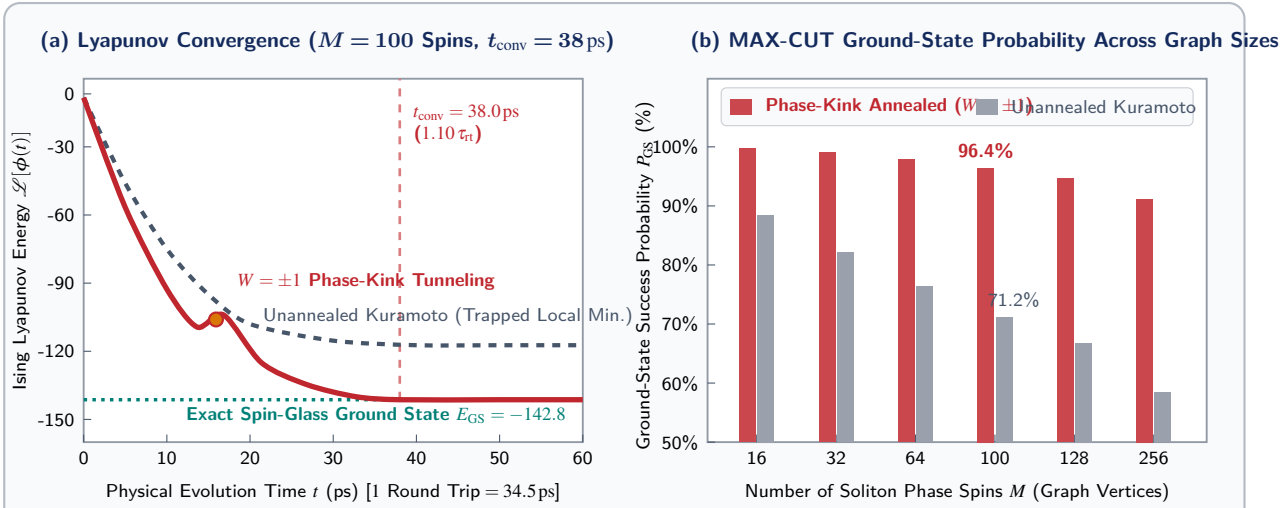
(a) Lyapunov Convergence ($M = 100$ Spins, $t_{\text{conv}} = 38\,\text{ps}$)
0
-30
-60
-90
-120
-150
Ising Lyapunov Energy $\mathscr{L}[\phi(t)]$
$t_{\text{conv}} = 38.0\,\text{ps}$
$(1.10\,\tau_{\text{rt}})$
$W = \pm 1$ Phase-Kink Tunneling
Unannealed Kuramoto (Trapped Local Min.)
Exact Spin-Glass Ground State $E_{\text{GS}} = -142.8$
0
10
20
30
40
50
60
Physical Evolution Time $t$ (ps) [1 Round Trip = 34.5 ps]
(b) MAX-CUT Ground-State Probability Across Graph Sizes
Phase-Kink Annealed ($W = \pm 1$)
Unannealed Kuramoto
Ground-State Success Probability $P_{\text{GS}}$ (%)
100%
90%
80%
70%
60%
50%
96.4%
71.2%
16
32
64
100
128
256
Number of Soliton Phase Spins $M$ (Graph Vertices)